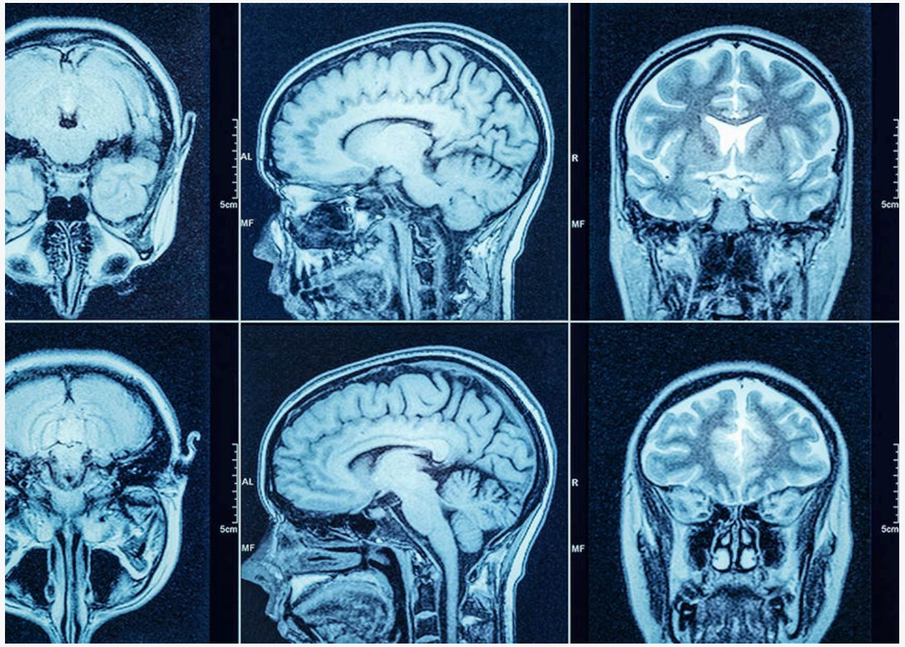
CT SCAN-BASED INJURY DETECTION AND CLASSIFICATION SYSTEM

DEEP LEARNING

CT (computerized tomography) Scans - The Problem



X-ray imaging that uses computer processing to create detailed cross-sectional body images from multiple angles. Diagnostic tool to identify diseases, injuries - in our case, brain injuries.

Some **statistics regarding diagnostic error** and misdiagnosis-**related harm rates**, highlighting the **variability in the accuracy of medical diagnoses** and the **impact** these errors can have on patient health:

21.9%
Emergency Department Revisits (with repeat CT or MRI)

13.6%
Diagnostic error rate (Median across various conditions)

11.5%
Radiologic Errors (of total diagnostic errors)

5.5%
Misdiagnosis-related harm rate

Need for efficient and accurate analysis of head CT scans in hospitals.
By attempting to detect the 6 brain injuries (ICH, IPH, IVH, SDH, EDH, SAH) with a high accuracy, our project aims to address this issue.

1. Lvmc. (2021, December 4). 12 reasons you may need a CT scan - Lompoc Valley Medical Center. Lompoc Valley Medical Center. https://www.lompocvmc.com/blogs/2021/december/12-reasons-you-may-need-a-ct-scan/
2. Newman-Toker, D. E., Wang, Z., Zhu, Y., Nassery, N., Tehrani, A. S. S., Schaffer, A. C., Yu-Moe, C. W., Clemens, G., Fanai, M., & Siegal, D. (2020). Rate of diagnostic errors and serious misdiagnosis-related harms for major vascular events, infections, and cancers: toward a national incidence estimate using the “Big Three.” Diagnosis, 8(1), 67–84. https://doi.org/10.1515/dx-2019-0104
3. Ahn, Y., Hong, G., Park, K. J., Lee, C. W., Lee, J. H., & Kim, S. (2021). Impact of diagnostic errors on adverse outcomes: learning from emergency department revisits with repeat CT or MRI. Insights Into Imaging, 12(1). https://doi.org/10.1186/s13244-021-01108-0

Literature Review

Automated and semi automated detection of findings from head CT scans have been studied by many groups.

- A study by Mansour and Aljehane employed an Inception Network for ICH segmentation. They utilised Kapur's thresholding combined with Elephant Herd Optimization (EHO) to optimize segmentation over CT images.
- Wang et al. implemented an XGBoost model to predict out-comes for aneurysmal subarachnoid haemorrhage (SAH) patients.
- In the work by Karkkainen et al. , the authors propose a fully unsupervised algorithm that leverages mixture models for ICH segmentation, emphasising the importance of preprocessing in their methodology.
- Mohammed et al. explored various systems for ICH detection by integrating low-dimensional vectors extracted from CT images using Gray Level Co-occurrence Matrix (GLCM) and Local Binary Pattern (LBP) with high-level features from models like AlexNet and ResNet-50.
- Liu et al. introduced a hybrid CNN-RNN architecture for ICH detection but faced complexities during training that led to vanishing gradient problems.
- The study by Rau et al. highlighted issues related to traditional machine learning classifiers like decision trees, which are prone to outliers and overfitting.

Sources

R.F. Mansour and N.O. Aljehane. An optimal segmentation with deep learning based inception network model for intracranial hemorrhage diagnosis. 33:13831–13843, 2021.
R. Wang, J. Zhang, B. Shan, M. He, and J. Xu. Xgboost machine learning algorithm for prediction of outcome in aneurysmal subarachnoid hemorrhage.18:659–667, 2022
Kimmo Karkkainen, Shayan Fazeli, and Majid Sarrafzadeh. Unsupervised acute intracranial hemorrhage segmentation with mixture models. arXiv preprint arXiv:2105.05891, 2021.
B.A. Mohammed, E.M. Senan, Z.G. Al-Mekhlafi, T.H. Rassem, N.M. Makbol, A.A. Alanazi,T.S. Almurayziq, F.A. Ghaleb, and A.A. Sallam. Multi-method diagnosis of ct images for rapid detection of intracranial hemorrhages based on deep and hybrid learning. Electronics, 11:2460,2022.
P. Liu, G. Ning, L. Shi, Q. Zhou, and X. Chen. Fine-grained classification of intracranial haemorrhage subtypes in head ct scans. IET Computer Vision, 17:170–188, 2023.
C.-S. Rau, S.-C. Wu, P.-C. Chien, P.-J. Kuo, Y.-C. Chen, H.-Y. Hsieh, and C.-H. Hsieh. Prediction of mortality in patients with isolated traumatic subarachnoid hemorrhage using a decision tree classifier: A retrospective analysis based on a trauma registry system.14(12):1420, 2017.

Literature Review

Detection of Subarachnoid Hemorrhage Using CNN with Dynamic Factor and Wandering Strategy-Based Feature Selection

Preprocessing:

- **Data augmentation** techniques like **flipping, shifting, and random rotation** were applied to increase the dataset size and reduce overfitting
- **Image Segmentation**:
  - **Modified Region-Growing Method** was applied using intensity and orientation thresholds.
  - **Sobel Operator** was used to calculate gradient magnitude and direction for precise edge detection.
- **Feature Extraction:**
  - **GoogleNet**: Extracted high-level patterns and irregularities.
  - **VGG-16:** Captured fine-grained textures and details.
  - **ResNet50**: Extracted complex and abstract features using residual connections.
- **Feature Optimization:**
  - **DWSCSO Algorithm**: Combined Water Wave Dynamic Factor (WWDF) for adaptive search and Wandering Strategy based Sand Cat Swarm Optimization for robust exploration.

ML Models Used:

- ResNet50, GoogleNet, and VGG-16 to detect Subarachnoid Hemorrhages (SAH) in CT scans

1. Sengupta, J., Alzbutas, R., Iešmantas, T., Petkus, V., Barkauskienė, A., Ratkūnas, V., Lukoševičius, S., Preikšaitis, A., Lapinskienė, I., Šerpytis, M., Misiulis, E., Skarbalius, G., Navakas, R., & Džiugys, A. (2024). Detection of Subarachnoid Hemorrhage Using CNN with Dynamic Factor and Wandering Strategy-Based Feature Selection. Diagnostics, 14(21), 2417. https://doi.org/10.3390/diagnostics14212417

Our Data

Digital Imaging and Communications in Medicine - **DICOM** data from **qure.ai**

This is a medical imaging format which has a lot of information about the patient, machines, doctor, etc.

Data was collected from Centre for Advanced Research in Imaging, Neurosciences and Genomics (CARING), New Delhi, India.

Our dataset has **472 CT scans, 21547 slices**.

- Each scan has multiple slices, each of which is a .dcm file.
- We use the **image information in the dcm file** + the **labels in a csv file.**

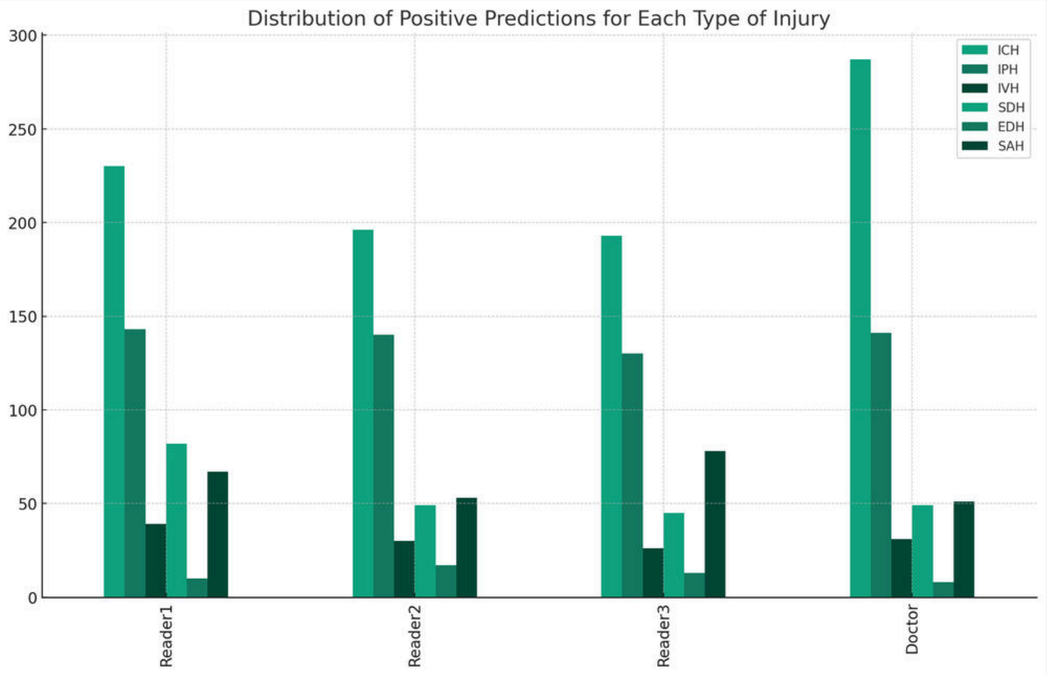
Ethical concerns were addressed by removing all identifiers from them.

- To **visualise the pixel values**, we plotted the slices in a scan to see the distribution. **HU units**, density ordered
  - -1000 is air
  - 0 is water
  - 1000 is bone.



1. Chilamkurthy, S., Ghosh, R., Tanamala, S., Biviji, M., Campeau, N. G., Venugopal, V. K., Mahajan, V., Rao, P., & Warier, P. (2018, March 13). Development and validation of deep learning algorithms for detection of critical findings in head CT scans. arXiv.org. https://arxiv.org/abs/1803.05854

True Labels - Bias & Weighting



	BleedLocation-Left	BleedLocation-Right	CalvarialFracture	ChronicBleed	EDH	Fracture	ICH	IPH	IVH
0	0.833333	1.000000	0.0	0.000000	0.000000	0.000000	1.000000	1.000000	0.000000
1	1.000000	0.000000	1.0	0.000000	0.833333	1.000000	1.000000	0.166667	0.000000
2	0.000000	0.000000	0.0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
3	0.000000	0.000000	0.0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
4	0.000000	0.000000	0.0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
5	0.000000	0.000000	0.0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
6	1.000000	0.166667	0.0	0.000000	0.000000	0.000000	1.000000	1.000000	1.000000
7	0.000000	0.000000	0.0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
8	0.000000	0.000000	0.0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
9	1.000000	1.000000	0.0	0.500000	0.000000	0.000000	0.666667	0.000000	0.000000
10	0.000000	0.000000	0.0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
11	1.000000	1.000000	0.0	0.000000	0.000000	0.000000	1.000000	1.000000	1.000000
12	0.166667	1.000000	0.0	0.000000	0.000000	0.000000	1.000000	1.000000	1.000000
13	0.666667	1.000000	0.0	0.000000	0.000000	0.166667	1.000000	1.000000	0.000000
14	1.000000	0.000000	0.0	0.500000	0.000000	0.000000	1.000000	0.000000	0.000000
15	0.000000	1.000000	0.0	0.000000	0.000000	0.000000	1.000000	1.000000	0.000000
16	0.000000	0.500000	0.0	0.166667	0.000000	0.000000	0.500000	0.500000	0.000000
17	0.000000	0.000000	0.0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
18	0.500000	1.000000	0.0	0.000000	0.000000	0.000000	1.000000	1.000000	0.166667
19	1.000000	0.166667	0.0	0.000000	0.000000	0.000000	1.000000	1.000000	0.000000

These patterns highlight the **individual tendencies** of each reader **towards** certain types of **injuries**.
Reader 1 - more aggressive, especially with ICH & IPH,
Reader 2 - focus on Midline Shift
Reader 3 - most conservative, close attention to ICH, Mass Effect.

Intracerebral Hemorrhage (ICH), Intraparenchymal Hemorrhage (IPH)
Intraventricular Hemorrhage (IVH), Subdural Hematoma (SDH)
Epidural Hematoma (EDH) , Subarachnoid Hemorrhage (SAH)

(a) Weighed the readers by experience,

- Reader 1 - weight 1 (8 years of experience),
- Reader 2 - weight 2 (12 years of experience),
- Reader 3 - weight 3 (20 years of experience)

The doctor's predictions were used as a reference for comparison but were not directly weighted.
(b) Summed their predictions grouped by injury,
(c) Normalised it (so the values earlier from 0 to 6 are now 0-1)

ResNet-18

ResNet (Residual Neural Networks)

- The key idea behind ResNets is **residual learning**. Instead of trying to learn the exact mapping from input to output, ResNets learn the "residual" or the **difference between the input and the desired output**. This makes it easier for the network to learn complex mappings and prevents degradation in performance as the network gets deeper.
- ResNet-18 has 18 layers (17 convolutional and 1 fully connected). This makes it **faster to train** and less memory intensive while still benefiting from deep feature learning.
- In medical imaging tasks, ResNet architectures, such as **ResNet-18**, **ResNet-50**, etc., are commonly used for their **ability to handle complex features** and learn representations from medical images effectively.

1. Bangar, S. (2022, July 5). Resnet Architecture Explained - Siddhesh Bangar - Medium https://medium.com/@siddheshb008/resnet-architecture-explained-47309ea9283d
2. He, K., Zhang, X., Ren, S., & Sun, J. (2015, December 10). Deep residual learning for image recognition. arXiv.org. https://arxiv.org/abs/1512.03385

GoogleNet

GoogleNet (Inception Network)

- GoogleNet is built on the concept of the **inception module**, which performs convolution operations of different sizes in parallel. This allows the network to capture both fine and coarse features within the same layer.
- GoogleNet consists of 22 layers (including convolutional, pooling, and fully connected layers) but has fewer parameters compared to traditional CNNs due to its efficient architecture. Its modular design makes it **computationally efficient despite being deep.**
- In medical imaging tasks, GoogleNet is frequently used because of its **ability to learn multi-scale features simultaneously**.

1. Bangar, S. (2024, December 10). GoogleNet: A deep dive into Google's neural network technology -Siddhesh Bangar - Medium. https://medium.com/@siddheshb008/googlenet-a-deep-dive-into-googles-neural-network-technology-f588d1b49e55
2. Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., & Rabinovich, A. (2015). Going deeper with convolutions. arXiv. https://arxiv.org/abs/1409.4842

Methodology

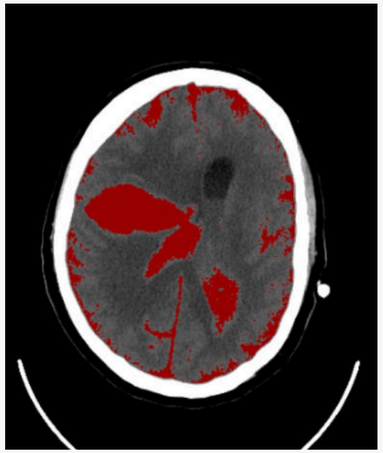
Preprocessing:

1. **Rescaling - so that we can do windowing**
2. **Windowing - below 0 converted to 0, above 100 converted to hundred**
3. **Removing skull outline - remove all bone recognised (intensity 100)**



Growing region Segmentation:

1. **Regions recursively grow if similarity criteria is matched, starting with an initial seed pixel for each segment.**
2. **The size of the region grows as the neighbouring pixels are added to the cluster of seed pixels**
3. **When the growth of one region stops, we choose another seed pixel which does not belong to any other region yet and repeat the same process**



4.a. Finetune ResNet 18

- **Modify the input to have 4 channels**
- **The final Fully connected layer to predict for 6 classes**

4b. Fintune GoogleNet

- **Modify the input to have 4 channels**
- **The final Fully connected layer to predict for 6 classes**

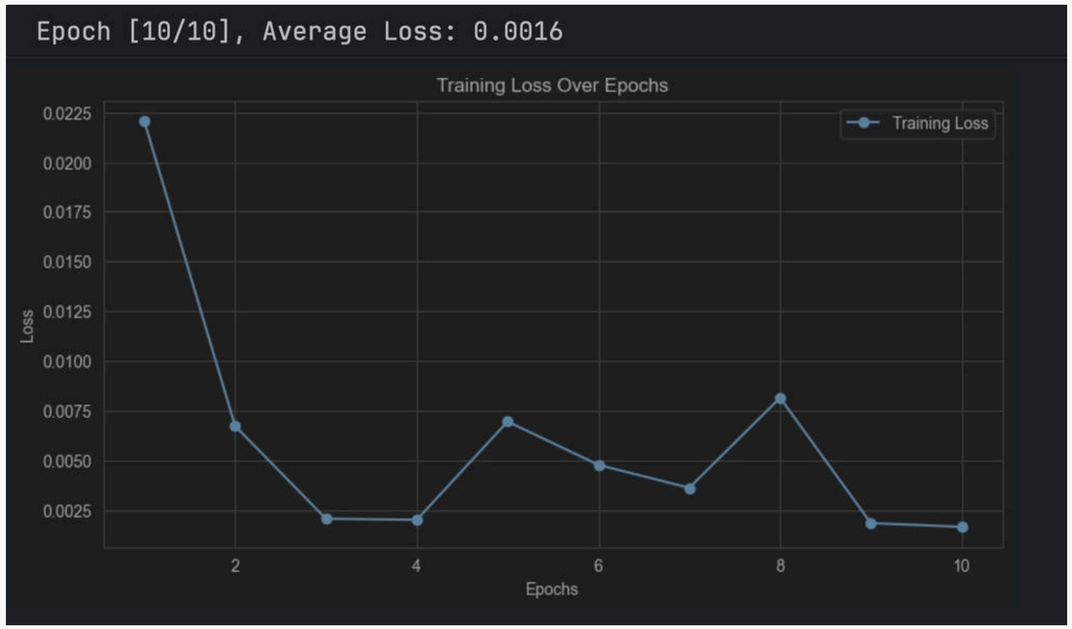
5. DWSCSO Optimiser

1. **Dynamic Water Wave Sand Cat Swarm Optimisation**
2. **optimises the features extracted from the finetuned resnet and googlenet**
3. **It has 2 phases exploration and exploitation**
4. **A group of candidate solutions is initialised**
    a. **Each candidate is evaluated using a fitness function**
5. **The exploration phase is responsible for searching the global solution space**
    a. **Starts of with a max value of 2 and gradually decreases**
6. **The exploitation search refines the search near promising solutions**
    a. **uses Levy flight to avoid local minima**

6. PRSCNN

1. **Is a simple CNN architecture with 2 1D convolutional layers and 2 FC layers**
2. **Each layer is followed by 1 PRelu function**
3. **The input is the optimised features selected by DWSCSO**
4. **And the output is the classification**

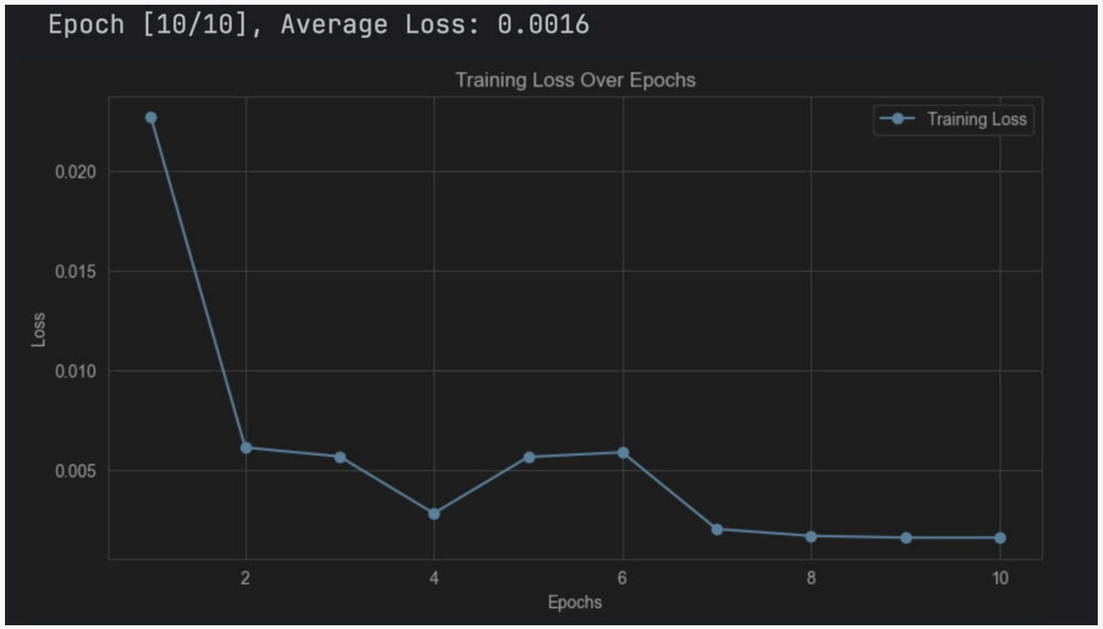
Results (ResNet18, GoogleNet)



Accuracy on Test Set: 99.72%

Average F1 Score: 0.8328205128205127

ResNet18



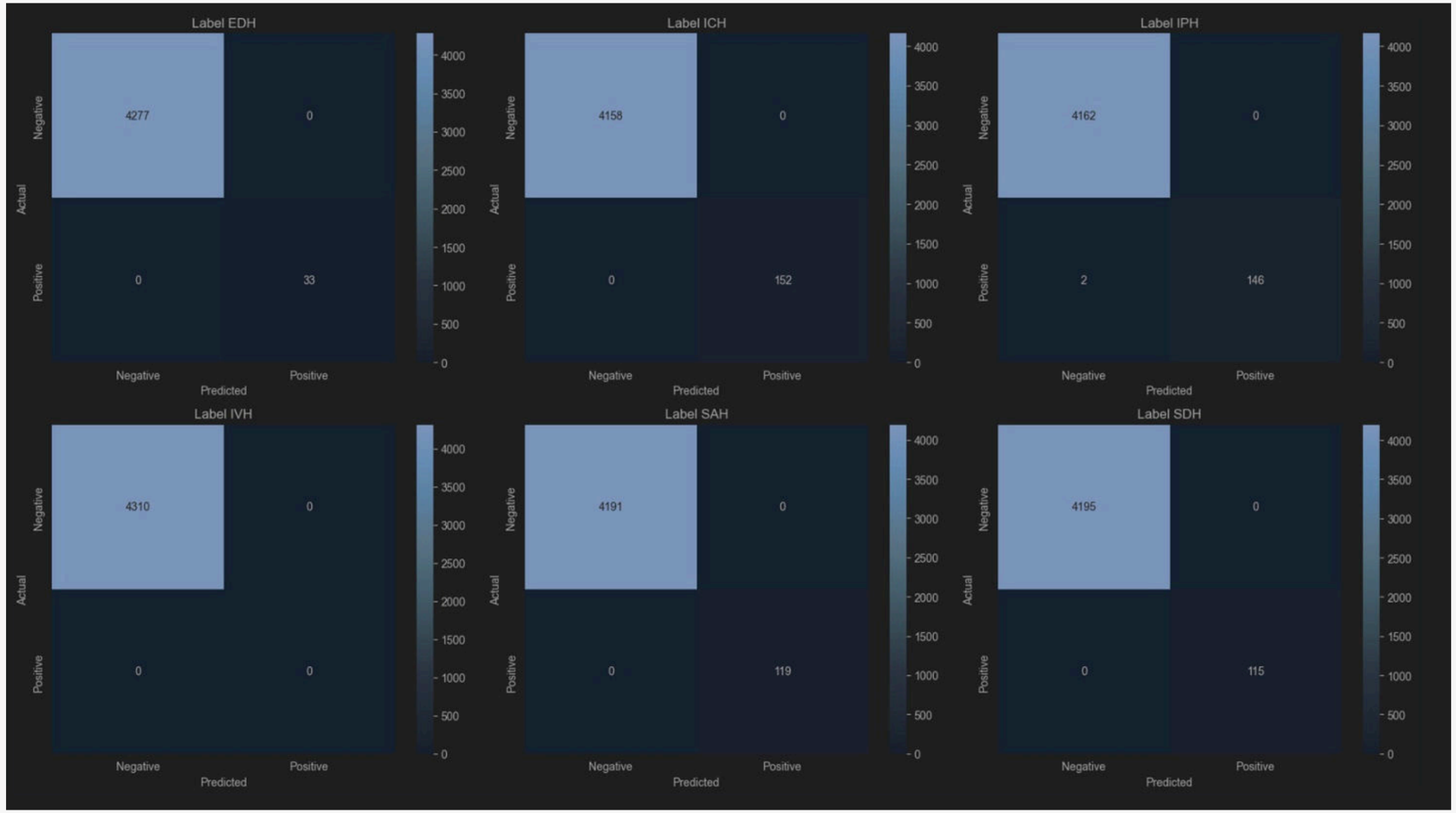
Accuracy on Test Set: 99.78%

Average F1 Score: 0.8321839080459771

GoogleNet

Final Results

(Preprocessing --> Segmentation --> ResNet18 + GoogleNet--> DWSCSO --> PRSCNN)



Accuracy on Test Set: 99.74%

Average F1 Score: 0.9986

2 slices were incorrectly predicted, our false positive rate is 0, false negative rate

Balanced Accuracy & Components

Specificity and Sensitivity

$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}}$$

$$\text{Sensitivity} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

$$\text{Balanced Accuracy} = \frac{1}{2}(\text{Sensitivity} + \text{Specificity})$$

$$\text{Average Balanced Accuracy} = \frac{1}{N}\sum_{i=1}^{N}\text{Balanced Accuracy}_i$$

Name of the injury	Sensitivity	Specificity	Balanced Accuracy
EDH	1.0	1.0	1.0
ICH	1.0	1.0	1.0
IPH	0.9865	1.0	0.9933
IVH	0.0	1.0	0.5
SAH	1.0	1.0	1.0
SDH	1.0	1.0	1.0

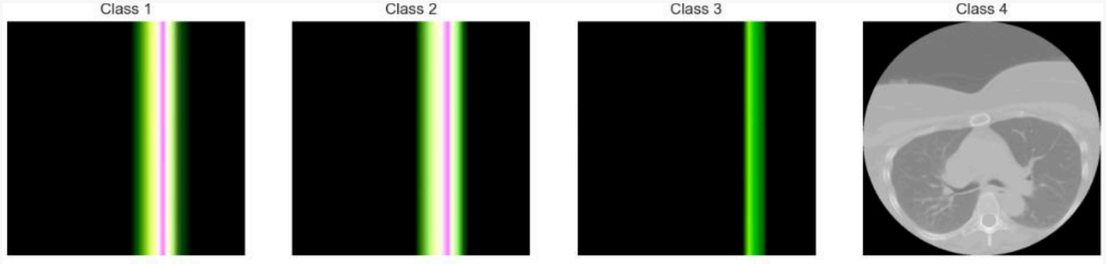
Average Balanced Accuracy : 0.9156 = **91.56%**

If the balanced accuracy < test accuracy (**99.74%**), it suggests that the model might not perform equally well across all classes, and that the dataset is skewed.

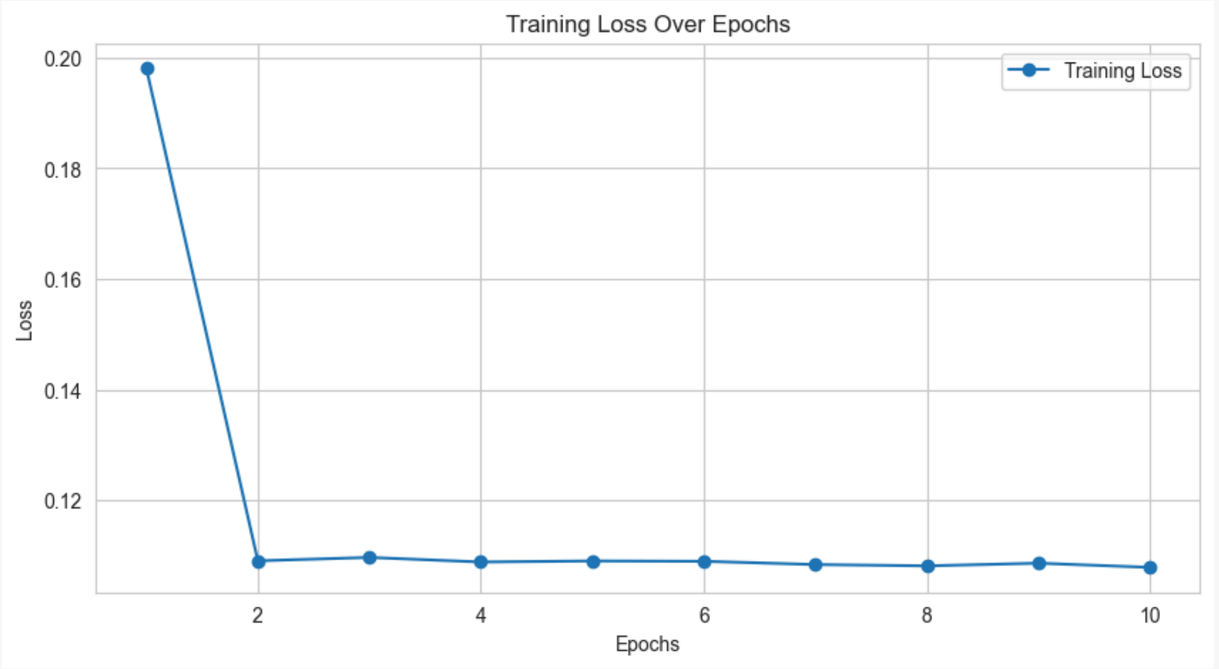
MUSEUM OF FAILURES

What we learned from trying

1. GradCAM
2. VGG16
3. Supervised learning

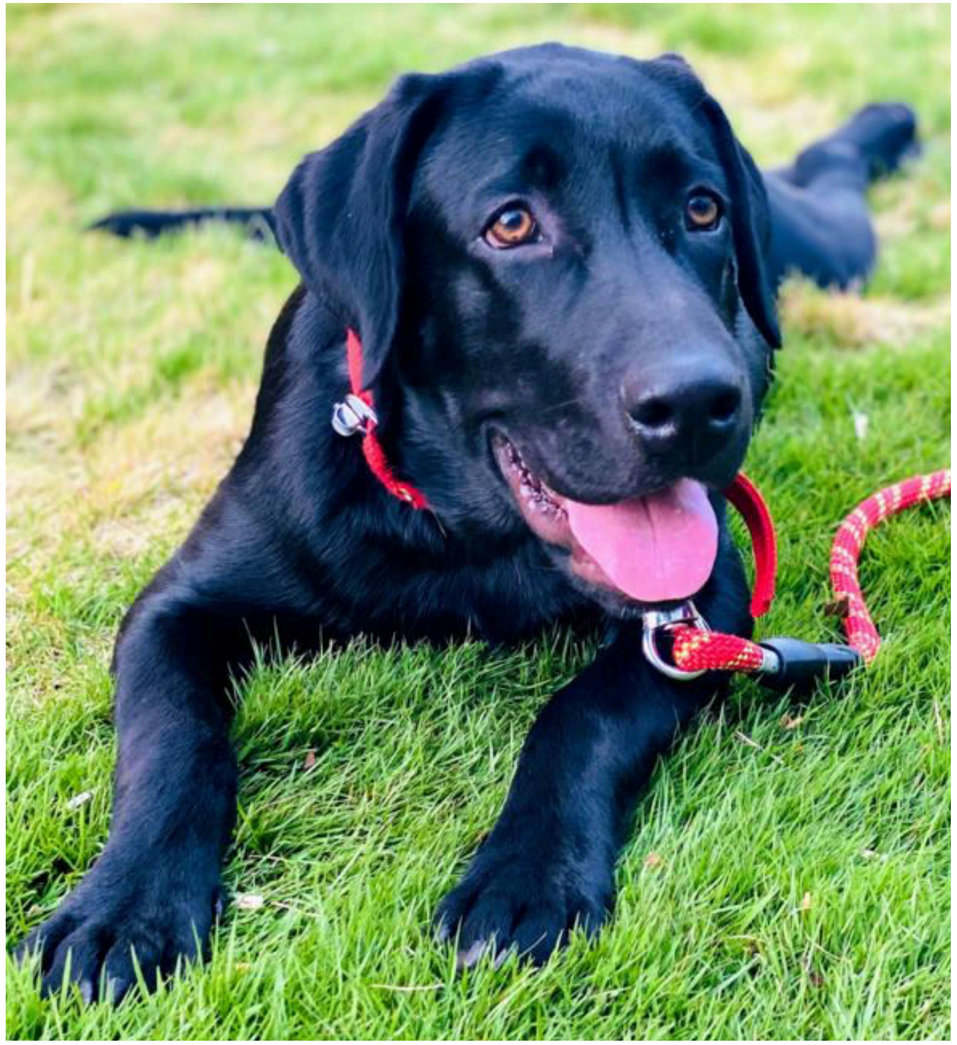


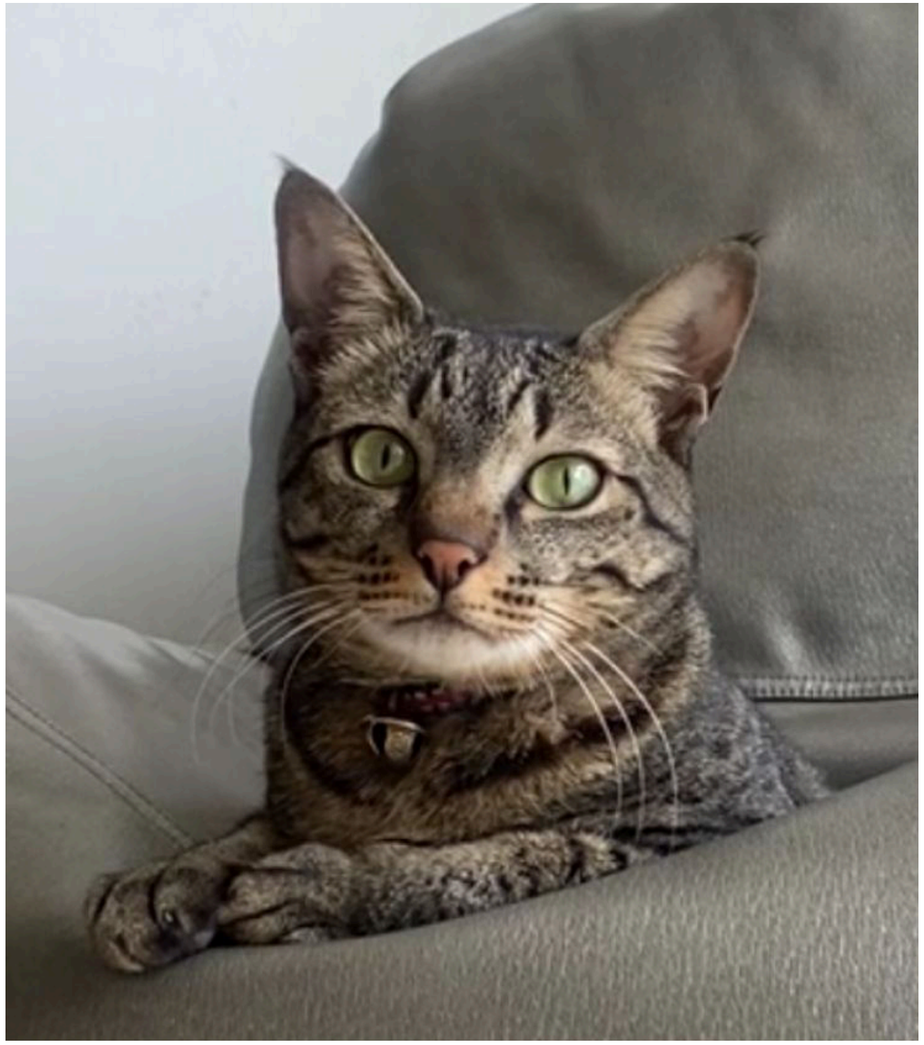
GradCAM “Results”




```
F1 Scores per label: [0. 0. 0. 0. 0. 0.]
Average F1 Score: 0.0
```

VGG16 “Results”





THANK YOU :)

PPT BY ACE & SAEBU

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = \frac{\text{TP}}{\text{TP} + \frac{1}{2}(\text{FP} + \text{FN})}$$

TP = number of true positives

FP = number of false positives

FN = number of false negatives